

RED OAK, IA, USA

WMO: 725494

Lat: 41.010N

Lon: 95.260W

Elev: 1044

StdP: 14.15

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04966

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.7	1.5	-11.0	3.2	-1.4	-6.4	4.1	4.8	27.5	24.8	25.2	24.8	4.6	0	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	94.8	77.7	91.3	76.5	90.0	75.9	81.4	90.8	79.5	88.8	77.6	86.6	11.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	156.7	88.1	76.9	145.5	86.1	74.8	135.4	84.1	46.1	90.8	43.7	88.2	41.9	86.4	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	21.7	19.1	DB	-13.1	97.9	6.3	2.8	-17.6	100.0	-21.3	101.6	-24.9	103.2	-29.5	105.3
			WB	-13.1	83.4	6.7	2.0	-17.9	84.8	-21.8	85.9	-25.6	87.0	-30.5	88.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.2	25.1	29.0	40.8	52.6	63.5	73.1	76.6	74.0	66.5	54.1	40.7	29.1
	DBStd	20.46	12.15	12.46	11.91	9.98	8.55	6.36	5.75	5.85	8.42	9.51	10.63	10.88
	HDD50	2817	772	591	333	85	6	0	0	2	65	312	651	
	HDD65	5820	1237	1007	754	388	132	10	2	5	83	357	731	1114
	CDD50	3624	0	4	47	162	425	693	825	745	496	191	34	2
	CDD65	1155	0	0	3	15	86	253	362	286	128	19	3	0
	CDH74	12262	0	0	35	237	954	2650	3923	2788	1379	237	59	0
	CDH80	4959	0	0	4	67	329	1076	1740	1129	519	62	33	0
Wind	WSAvg	8.0	8.3	8.6	9.7	10.6	9.4	7.8	5.7	5.1	6.4	7.8	8.3	7.8
Precipitation	PrecAvg	35.7	0.9	1.1	2.2	3.7	5.3	5.6	4.6	3.9	3.0	2.6	1.7	1.0
	PrecMax	52.6	1.9	2.4	4.6	8.6	10.2	13.5	11.7	9.0	7.3	8.7	4.5	2.8
	PrecMin	25.1	0.2	0.2	0.1	0.8	1.5	1.3	1.5	0.6	0.7	0.5	0.0	0.0
	PrecStd	7.4	0.4	0.6	1.4	1.7	2.4	3.3	2.4	2.3	1.7	2.0	1.1	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.6	66.0	78.9	88.1	93.5	95.1	99.3	97.2	94.7	86.3	80.7	62.7
		MCWB	45.9	54.0	60.1	65.1	71.1	76.2	79.5	78.3	74.6	68.9	64.3	54.8
	2%	DB	52.0	58.6	72.3	81.1	88.0	91.5	95.0	92.7	90.2	80.9	69.7	54.4
		MCWB	44.1	49.2	57.9	62.7	70.1	75.2	79.3	78.7	73.9	64.4	56.1	47.3
	5%	DB	45.7	52.0	65.8	75.2	82.4	90.0	91.3	90.0	86.1	75.0	63.0	49.7
		MCWB	40.1	44.7	54.8	59.3	67.0	74.2	78.1	78.2	72.7	62.4	52.8	44.3
	10%	DB	41.0	45.7	60.6	70.4	79.3	86.3	89.8	86.2	81.9	71.5	57.2	44.8
		MCWB	36.5	40.5	50.9	57.0	65.4	72.7	77.6	75.6	69.1	59.5	49.3	39.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.8	56.2	64.9	68.6	76.3	80.4	84.2	84.1	79.1	73.2	64.7	57.5
		MCDB	54.9	62.3	74.0	80.7	87.6	90.8	93.2	92.8	89.5	81.2	75.4	61.7
	2%	WB	44.1	49.7	60.4	65.2	72.6	78.0	81.7	81.2	76.4	68.2	58.9	49.0
		MCDB	50.7	58.4	68.1	76.9	83.1	88.2	90.9	90.2	86.3	75.2	65.9	52.7
	5%	WB	40.5	44.9	55.7	62.0	70.1	76.2	79.9	78.9	74.3	64.5	54.4	43.6
		MCDB	45.9	51.1	64.0	72.8	80.0	86.4	89.4	87.4	83.1	71.3	62.0	49.0
	10%	WB	36.9	40.8	50.9	58.9	67.4	74.3	78.2	76.9	71.8	60.8	49.8	39.5
		MCDB	40.9	45.8	59.8	68.2	76.4	84.3	87.1	84.7	79.7	68.9	57.2	44.0
Mean Daily Temperature Range	5% DB	MDBR	18.6	19.0	21.5	23.4	21.6	20.8	20.6	21.1	24.7	23.7	21.6	17.9
		MCDBR	26.3	27.0	31.3	31.7	26.6	23.4	22.9	23.9	27.0	29.5	30.4	25.5
		MCWBR	19.6	18.5	18.5	17.2	12.9	11.1	12.1	13.5	14.0	16.1	19.0	18.9
	5% WB	MCDBR	25.0	25.3	27.2	26.1	22.3	20.5	20.3	21.2	23.5	23.6	27.5	23.0
	MCWBR	19.0	18.4	17.5	16.3	12.2	11.9	12.5	13.3	13.3	15.1	19.1	17.8	
Clear-Sky Solar Irradiance	taub	0.282	0.313	0.327	0.384	0.409	0.414	0.429	0.419	0.379	0.334	0.296	0.285	
	taud	2.450	2.379	2.405	2.265	2.264	2.300	2.248	2.292	2.359	2.472	2.508	2.455	
	Ebn at Noon	280	285	293	281	276	273	268	267	272	274	273	269	
	Edh at Noon	24	30	33	41	43	41	43	40	35	28	23	23	
All-Sky Solar Radiation	RadAvg	643	939	1216	1540	1831	2031	2044	1776	1477	1025	710	540	
	RadStd	53	60	111	103	146	166	143	107	99	120	55	54	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page